Adverse Drug Reactions & Pharmacovigilance

Exam-oriented notes based on the DigiNerve lecture by Dr Nirmal George (30 April 2025), supplemented with standard pharmacovigilance terminology.

1. Adverse Drug Reaction (ADR)

WHO definition: A noxious, unintended response to a medicine occurring at doses normally used in humans for prophylaxis, diagnosis, therapy, or modification of physiological function.

Adverse Event (AE) ≠ ADR: An AE is any untoward medical occurrence after exposure to a medicine; a causal relationship with the medicine does not have to be established.

Key point: Suspected ADRs should be reported even when causality is not yet proven.

2. Classification of ADRs

Type	Meaning / key feature	Typical example
Pharmacological	Predictable from the drug's known pharmacology; often dose-related.	Hypoglycaemia with insulin
Pathological	Produces or worsens a pathological condition.	Drug-induced organ injury
Genotoxic	Causes genetic/DNA damage.	Certain cytotoxic agents
Allergic	Immune-mediated reaction; requires sensitization.	Penicillin hypersensitivity
Idiosyncratic	Unpredictable, unusual reaction not explained by usual pharmacology.	Unusual drug-induced reaction
Pharmacogenetic	Related to inherited genetic differences in drug response.	G6PD-related haemolysis
Teratogenic	Causes fetal developmental abnormalities when exposure occurs during pregnancy.	Thalidomide

3. Hypersensitivity Reactions (Types I–IV)

Type	Mechanism	Common examples
I – Immediate	IgE-mediated; mast-cell degranulation	Anaphylaxis, urticaria
II – Cytotoxic	IgG/IgM against cell-surface antigens → cell injury	Drug-induced haemolytic anaemia
III – Immune complex	Antigen–antibody complexes deposit in tissues	Serum sickness
IV – Delayed	T-cell mediated	Contact dermatitis

4. Teratogenicity & Pregnancy Drug Categories

Exam note: The lecture discusses the historical FDA A/B/C/D/X categories. The US FDA replaced these letter categories with the Pregnancy and Lactation Labeling Rule (PLLR) for prescription drug labeling; therefore, A–X should be treated as a traditional/exam framework, not a current FDA labeling system.

Category	Traditional interpretation
A	Controlled human studies show no risk to fetus.
B	No evidence of risk in humans; animal data reassuring or human data limited.
C	Risk cannot be ruled out; use only when potential benefit justifies risk.
D	Positive evidence of fetal risk; may be used when benefits outweigh risks.
X	Fetal risk clearly outweighs any possible benefit; contraindicated in pregnancy.

5. Causality Assessment of Suspected ADRs

Purpose: Estimate how likely it is that a medicine caused the observed event. Causality assessment supports clinical evaluation and signal detection; it does not prove causation with certainty.

Tool	Core idea / use
WHO–UMC scale	Clinical/global assessment using factors such as temporal relationship, alternative causes, dechallenge/rechal
Naranjo algorithm	Structured questionnaire producing a score that categorizes causality (e.g., definite, probable, possible, doubt
Bradford Hill considerations	General causal-inference considerations such as temporality, strength, consistency, biological plausibility and

6. Pharmacovigilance (PV)

WHO: The science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other medicine-related problem.

Major objectives: detect new safety signals • characterize known risks • improve safe use of medicines • support regulatory action • reduce preventable medication-related harm.

Scope: includes medicines, vaccines, biological products and other medicine-related safety problems; PV extends beyond spontaneous ADRs to broader safety surveillance.

7. ADR Reporting in India – PvPI

Pharmacovigilance Programme of India (PvPI) is India's national drug-safety monitoring programme. It was initiated in 2010 and the National Coordination Centre was shifted to the Indian Pharmacopoeia Commission (IPC), Ghaziabad, in 2011.

Who can report? Healthcare professionals such as doctors, dentists, pharmacists and nurses, as well as patients/consumers.

What to report? Suspected ADRs, including serious/non-serious, known/unknown and frequent/rare reactions. You do not need to wait for proof of causality before reporting.

8. Practical ADR Reporting Workflow

1	**Identify**	Recognize a suspected adverse reaction in a patient.
2	**Collect**	Record patient details, suspected medicine(s), dose, route, dates, reaction, outcome and relevant history.
3	**Assess**	Consider seriousness, temporal relationship, alternative causes and dechallenge/rechallenge where applicable
4	**Report**	Complete the suspected ADR reporting form / use the appropriate PvPI reporting route.
5	**Follow up**	Provide additional clinical information when available and document the outcome.

9. High-Yield Exam Points

• **ADR:** noxious + unintended response occurring at normally used doses.

• **AE:** temporal association with a medicine does not necessarily imply causality.

• **Hypersensitivity:** Type I = IgE; Type II = antibody/cell injury; Type III = immune complexes; Type IV = T cells.

• **Causality:** WHO–UMC and Naranjo are commonly used ADR causality approaches.

• **Reporting principle:** suspicion is sufficient to report; causality need not be established first.

• **PvPI:** India's national pharmacovigilance programme; NCC is at IPC, Ghaziabad.

Primary video: DigiNerve, “Understanding Adverse Drug Reactions & Pharmacovigilance,” Dr Nirmal George, published 30 April 2025. The public video description lists ADR definition/classification, hypersensitivity, pregnancy categories, Bradford Hill/WHO–UMC/Naranjo causality assessment, ADR reporting, PvPI and CDSCO reporting tools.

Reference: Indian Pharmacopoeia Commission (IPC), Pharmacovigilance Programme of India (PvPI) FAQ and programme information.